

Question

Execute the command `cat names.txt | grep "Alice"` using an anonymous pipe and understand how data flows between the `cat` and `grep` processes

```

GNU nano 2.9.3          names.txt
Alice
John
David
Alice
Robert
Mary

[?] Help      [?] Write Out  [?] Where Is   [?] Cut        [?] Execute   [?] Location  [?] Undo      [?] Set Mark  [?] To Bracket
[?] Exit      [?] Read File  [?] Replace   [?] Paste      [?] Justify   [?] Go To Line [?] Redo      [?] Copy      [?] Where Was

Child process: 5
bhavan@LAPTOP-FGDPB110:~$ vmstat 1 5
procs  -----memory-----  --swap--  ---io---  -system-  -----cpu-----
r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs  us  sy  id  wa  st  gu
1  0     0  7356360  5212  196640   0   0    800    17   261    0  0  0  100  0  0  0
0  0     0  7356360  5212  196640   0   0     0     0   176   68  0  0  100  0  0  0
0  0     0  7356360  5212  196712   0   0     0     0   159   64  0  0  100  0  0  0
0  0     0  7356360  5212  196712   0   0     0     0    73   36  0  0  100  0  0  0
0  0     0  7356360  5212  196712   0   0     0     0    77   36  0  0  100  0  0  0
bhavan@LAPTOP-FGDPB110:~$ chrt -p $$
pid 329's current scheduling policy: SCHED_OTHER
pid 329's current scheduling priority: 0
bhavan@LAPTOP-FGDPB110:~$ nano names.txt
bhavan@LAPTOP-FGDPB110:~$ cat names.txt
Alice
John
David
Alice
Robert
Mary
bhavan@LAPTOP-FGDPB110:~$ cat names.txt | grep "Alice"
Alice
Alice
bhavan@LAPTOP-FGDPB110:~$ |

```

Short Theory

An **anonymous pipe** is used for communication between processes. In this command, cat reads the contents of names.txt and sends its output through the pipe |. The grep command receives this output and displays only the lines containing **"Alice"**.

The data flow is:

names.txt → cat → pipe → grep → Alice

Important points: pipe() creates a pipe, fd[0] is the read end, fd[1] is the write end, and dup2() can be used in C programs to redirect input and output through a pipe.

